Primer Evaluación Parcial (Tarea-Examen)

Nombre: _____

Matrícula: _____

Instrucciones:

Resolver 10 ejercicios, la siguiente lista de ejercicios, la solución debe de ser clara e incluir todos los pasos que llevaron a encontrar la respuesta correcta.

Obligatorios indicados con un asterísco – *

Pregunta 1. *Verifica una de las dos siguientes identidades*

(c) $B \setminus A = B \cap A^c$

(d) $A \cup B = A \cup (B \cap A^c)$

Pregunta 2. ** Sea S un espacio muestral. Muestre que la colección $\mathcal{B} = \{\emptyset, S\}$ es una sigma algebra.*

Pregunta 3. *Si A and B son mutuamente excluyentes, entonces*

$$P(A|A \cup B) = \frac{P(A)}{P(A) + P(B)}.$$

Pregunta 4. $P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$.

Pregunta 5. $P(A \cup B \cup C) = P(A) + P(A^c \cap B) + P(A^c \cap B^c \cap C)$.

Pregunta 6. *(Resolver el siguiente problema)*

When coded messages are sent, there are sometimes errors in transmission. In particular, Morse code uses "dots" and "dashes," which are known to occur in the proportion $3 : 4$. This means that for any given symbol, $P(\text{dot sent}) = \frac{3}{7}$ and $P(\text{dash sent}) = \frac{4}{7}$. Suppose there is interference on the transmission line, and with probability $\frac{1}{8}$ a dot is mistakenly received as a dash, and vice versa. If we receive a dot, can we be sure that a dot was sent?

Pregunta 7. ** For any three events A, B, and C defined on a sample space S, verify the following properties:* ***DeMorgan's Laws:*** $(A \cup B)^c = A^c \cap B^c$ *and* $(A \cap B)^c = A^c \cup B^c$

Pregunta 8. ** If A and B are two independent events defined on a given probability space $(\Omega, \mathcal{A}, P)$, then*

a) *A and B^c are independent,*

b) *A^c and B are independent,*

c) *A^c and B^c are independent.*

Pregunta 9. * *Assume the probability space $(\Omega, \mathcal{A}, P)$ is given and let $B \in \mathcal{A}$ satisfy $P(B) > 0$.*

a) $P(\emptyset|B) = 0$.

c) *If $A \in \mathcal{A}$ then $P(A^c|B) = 1 - P(A|B)$.*

e) $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1A_2|B)$.

f) *If $A_1 \subset A_2$, then $P(A_1|B) \leq P(A_2|B)$.*

Pregunta 10. *Properties of Indicator Functions: Let $\mathcal{A}$ be a collection of subsets of Ω.*

i) $I_{A^c}(\omega) = 1 - I_A(\omega)$.

ii) $I_{A_1 \cap A_2}(\omega) = I_{A_1}(\omega) I_{A_2}(\omega)$

Pregunta 11. *Solve the following*

a) *Show that $P(A|B) + P(A'|B) = 1$, provided $P(B) \neq 0$.*

b) *Show that $P(A \cap B \cap C \cap D) = P(A)\, P(B|A)\, P(C|A \cap B)\, P(D|A \cap B \cap C)$, provided $P(A \cap B \cap C) \neq 0$.*

Pregunta 12. *Show that if set B is contained in set A (that is, B is a subset of A), then $P(B) \leq P(A)$.*

Pregunta 13. *If A and B are mutually exclusive events, $P(A) = 0{,}33$, and $P(B) = 0{,}42$, find*

i) $P(A')$;

ii) $P(B')$;

iii) $P(A \cup B)$;

iv) $P(A \cap B)$;

v) $P(A \cap B')$;

vi) $P(A' \cap B')$.

Pregunta 14. *Show that*

i) $P(A \cap B) \leq P(A) + P(B)$;

ii) $P(A \cap B) \geq P(A) + P(B) - 1$.

Pregunta 15. *

d) *Show that*

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &+ P(A \cap B \cap C) \end{aligned}$$